

The FAT16 filesystem, introduced for MSDOS, had a 16-bit FAT counter. This meant it could write 65,535 entries into the FAT. Each entry corresponded to one block, and each block could hold 32 kilobytes of data. This added up to a support for only 2 GB disks. Then came FAT32, which could theoretically support disks up to 2 terabytes in size, but there have been problems with this, and to be safe, the maximum disk size is limited to 32 GB.

The NTFS (New Technology File System), introduced by Microsoft for Windows NT, took file organisation one step further. For one thing, it added *journaling*—it writes changes to data in a “journal” before it actually makes the changes. This way, if the system crashes and the OS sees data that doesn’t make sense, it just runs through the journal and keeps making changes till everything is nice and consistent once again. NTFS also brought security and access control to file storage.

Another type of filesystem is the *Log Structured File System*. In this case, the journal is the filesystem itself, which allows for “time travel”—you could go back and look at older versions of the same data. If data gets corrupted, it becomes easy to just go back to when everything was running fine. Right now, though, there is no Log Structured File System that has commercial popularity.

1.6 Summing It Up

The OS, we have seen, enters the picture everywhere—whether it’s juggling the various windows you have open at any time, or fetching data from your hard disk, or assigning memory to a program. Unless there’s a programmer willing to write code just for *your* hardware configuration, your computer—without an OS—can’t do much except blink lights on and off.